Current Affairs 09-11-2026

Credit - Drishti Ias

Linking Voter IDs with Welfare Benefits

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For Prelims: EPIC , Aadhaar , JAM Trinity , Direct Benefit Transfer , One Nation One Ration Card , Election Commission of India , Article 21.

For Mains: Balancing efficient welfare delivery with inclusivity and constitutional rights, Challenges faced by internal migrants and marginalised communities in accessing welfare benefits.

Source: TH

Why in News?

The Union Minister for Panchayati Raj proposed that **Voter Identity Cards (EPIC)** , coupled with biometric authentication, could replace or supplement **Aadhaar** as the primary document for identifying beneficiaries of state welfare schemes. However, this proposal has sparked a significant debate **regarding exclusion, administrative feasibility, and constitutional rights** .

Summary

- The proposal to link Voter IDs with welfare benefits raises concerns over constitutional rights, exclusion of vulnerable groups, migrant welfare portability and administrative feasibility.
- Welfare reforms should focus on technology-driven delivery, dynamic social registries and stronger DBT mechanisms, ensuring efficiency without compromising inclusivity and the Right to Food.

What is the Traditional Architecture of Welfare Delivery in India?

- **Aadhaar and Direct Benefit Transfer (DBT):** The cornerstone of modern Indian welfare is the **JAM Trinity (Jan Dhan Bank Accounts, Aadhaar, and Mobile**).
  - Aadhaar acts as a unique "financial address." By seeding Aadhaar with bank accounts, this architecture enables **Direct Benefit Transfer (DBT)** , bypassing middlemen to route subsidies (like **PM-KISAN**) directly into beneficiary accounts.

- **Public Distribution System (PDS) & Ration Cards:** Historically the foundational document for welfare, the ration card ensures national food security.
  - This system has been digitized into the **One Nation One Ration Card (ONORC)** framework, allowing **citizens to claim subsidized food grains anywhere in India using Aadhaar authentication.**
- **Purpose-Specific Cards:**
  - **MGNREGA Job Cards (now Gramin Rozgar Guarantee Cards under VB-G RAM G Act, 2025):** Issued at the Panchayat level to guarantee 100 days of wage employment.
  - **Ayushman Bharat (PM-JAY) Cards:** Issued for **universal health coverage** up to Rs 5 lakh per family.
  - **e-Shram Portal:** An initiative to create a national database of unorganized workers to deliver social security benefits.
- **State-Level Family Cards:** Moving away from individual targeting, states like **Rajasthan (Jan Aadhaar), Madhya Pradesh (Samagra), and Haryana (Parivar Pehchan Patra)** use household-level identity cards to streamline **family-centric welfare** (like housing and rations).

Current Status of Voter IDs in Welfare Delivery

- **As an "Entry Document":** In common parlance, the Voter ID is often used as a foundational proof of age and residence to obtain other primary welfare documents (e.g., applying for a **Gramin Rozgar Guarantee Card** **or opening a rural bank account).**
- **State-Specific Mandates:** In jurisdictions like Delhi, Voter IDs are mandatory for specific localized **pension schemes, often requiring endorsement from local legislators.**
- **No Exclusive National Mandate:** To date, Voter IDs have not been exclusively used as the primary financial or verification document for broad-based Central welfare schemes, **as they lack integration with the banking architecture.**

What are the Arguments For and Against Linking Voter IDs with Welfare Benefits?

Arguments For

- **Robust Verification Mechanism:** The **Shanta Kumar Committee (2015)** highlighted that massive inclusion and exclusion errors have historically plagued welfare delivery (like PDS).
  - The **Election Commission of India (ECI)** conducts rigorous, door-to-door physical verifications and continuous updates of electoral rolls.
  - Proponents argue this makes the **Voter ID less prone to the creation of "ghost beneficiaries" compared to Aadhaar, which is relatively easier to generate.**
- **Fiscal Prudence for States:** Linking welfare benefits with local electoral records may help states verify residency-based eligibility and detect duplicate claims, potentially improving the targeting and fiscal efficiency of state-specific welfare schemes.
- **Curbing Dual-State Dipping:** It prevents individuals from illegally claiming **localized welfare benefits** (like state pensions or housing schemes) in multiple states simultaneously.

Arguments Against

- **Violation of the Right to Life (Article 21):** In the landmark ***PUCL vs. Union of India (2001)*** "Right

to Food" case, the Supreme Court established that the Right to Food is an integral part of **Article 21.**

- The judgment strictly mandated that procedural barriers or the lack of specific identity documents (like a Voter ID) cannot be used by the State as an excuse to deny food to a starving population.

- **Article 14 & 19:** Denying welfare based on electoral status may restrict the **fundamental right to move and settle anywhere in India.**
- **Failing the Proportionality Test:** In the ***Justice K.S. Puttaswamy vs. Union of India (2018)*** Aadhaar verdict, the Supreme Court upheld Aadhaar for welfare delivery only because it passed a stringent "proportionality test" to prevent leakages.
  - Linking welfare to Voter IDs risks excluding internal migrants, minors and unregistered citizens, potentially failing the **proportionality test due to its exclusionary impact.**
- **Threat of Double Exclusion:** Linking welfare eligibility with electoral rolls undergoing **Special Intensive Revision (SIR)** could create **cascading exclusion** , where individuals removed from voter lists may also lose access to essential welfare benefits.
  - In **June 2026, West Bengal initiated verification of PDS beneficiaries against revised electoral rolls** , allowing ration card deactivation for ineligible individuals identified through SIR. However, beneficiaries with pending electoral appeals were provisionally protected, highlighting the need for **due process and continuity of welfare entitlements** .
- **Exclusion of Internal Migrants:** India has a massive circular migrant workforce. Migrants usually retain Voter IDs from their home states.
  - Mandating local Voter IDs for welfare effectively destroys the portability of welfare, completely undermining the **vision of the ONORC scheme.**
- **Lack of Financial Architecture:** The **Nandan Nilekani Task Force (2011)** laid the architecture for the modern JAM Trinity. The task force pushed for a centralized, biometric-backed financial address (Aadhaar) linked to core banking solutions, which Voter IDs fundamentally lack.
  - Unlike Aadhaar, which is mapped directly to the banking system (**NPCI** mapper) for DBT, **Voter IDs have no native financial linkage, making them administratively inefficient for processing payments.**

What Measures are Needed for Effective Welfare Delivery?

- **Programmable Money (CBDC and e-RUPI):** Scale up **e-RUPI** (programmable vouchers delivered to mobiles) and integrate into DBT.
  - This can be a potential solution for zero leakage and complete traceability without requiring complex localized documents.
- **Dynamic Social Registry:** Move away from static lists like SECC-2011. Establish a **National Social Registry** - a real-time database that updates socio-economic status based on verifiable life events, as recommended by the **Hashim Committee (2012)** .
- **Upgrading the JAM Architecture:** Address Aadhaar authentication failures (especially for manual laborers) by expanding touchless methods like **Iris scanning** and **Facial Recognition (FRT)** , alongside OTP fallbacks, adhering to the Supreme Court's mandate that authentication failure must not lead to denial of benefits.
- **Strengthening Welfare Portability:** Prioritize the full implementation of **ONORC** and integrate it with the **e-Shram portal** to ensure internal migrants can access benefits seamlessly across state borders.
- **Universalization & Social Audits:** Shift from rigid targeting to universal access for basic needs

(healthcare, elementary education). Empower Gram Sabhas to conduct mandatory, community-led **social audits** to verify beneficiaries organically.

- **Decouple Welfare Entitlements from Electoral Disputes:** Establish **independent beneficiary verification** and time-bound grievance redressal mechanisms so that an individual's pending electoral registration dispute does not automatically suspend essential benefits.
  - Adopt a **continuity-of-benefits principle** , allowing eligible individuals to establish identity through alternative documents while verification disputes are resolved.

Conclusion

Welfare reforms must balance efficiency with inclusivity and constitutional rights. Preventing leakages should not lead to exclusion of vulnerable groups. Technology-driven solutions must strengthen welfare delivery while ensuring that basic welfare remains a right, not a privilege linked to voter participation.

> ***Drishti Mains Question:***
>
> How can India balance beneficiary verification, fiscal prudence and welfare portability while designing an inclusive social protection architecture?

Frequently Asked Questions (FAQs)

1. What is the JAM Trinity and what is its significance in welfare delivery?
JAM stands for Jan Dhan, Aadhaar and Mobile. It facilitates financial inclusion, identity verification and Direct Benefit Transfer (DBT), reducing intermediaries and leakages.

2. What is the significance of the PUCL vs. Union of India (2001) judgment?
The Supreme Court recognised the Right to Food as an integral part of Article 21, strengthening the State's obligation to protect food security and prevent starvation.

3. What was the significance of the Justice K.S. Puttaswamy (2018) judgment on Aadhaar?
The Supreme Court upheld Aadhaar for targeted welfare subsidies subject to constitutional safeguards and the proportionality test, while restricting its mandatory use in several other contexts.

4. What is the One Nation One Ration Card (ONORC) scheme?
ONORC enables beneficiaries under the National Food Security Act (NFSA), 2013, to access subsidised food grains from any Fair Price Shop across India through portability.

5. What are e-RUPI and CBDC, and how can they improve welfare delivery?
e-RUPI is a prepaid, purpose-specific digital voucher, while CBDC (Digital Rupee) is the digital form of central bank money. Both offer potential for transparent and targeted digital payments.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/umYJLxMLZ9E**]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. Among the following who are eligible to benefit from the "Mahatma Gandhi National Rural Employment Guarantee Act"? (2011)

(a) Adult members of only the scheduled caste and scheduled tribe households
(b) Adult members of below poverty line (BPL) households
(c) Adult members of households of all backward communities
(d) Adult members of any household

Ans: (d)

Mains

Q. Performance of welfare schemes that are implemented for vulnerable sections is not so effective due to absence of their awareness and active involvement at all stages of policy process. Discuss. **(2019)**

Q. "Besides being a moral imperative of a Welfare State, primary health structure is a necessary precondition for sustainable development." Analyse. **(2021)**

Punjab-Haryana-Rajasthan Water Dispute

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Source: IE

Why in News?

Marking the **60 th anniversary of its reorganisation on Foundation Day (1 st November),** the Punjab Assembly passed five resolutions to revive decades-old **inter-state river water-sharing disputes.**

- The resolutions demanded an equitable share in **Yamuna waters** , Rs 1.44 lakh crore in royalty from Rajasthan, control over the Ropar, Harike and Ferozepur headworks, repeal of the **Dam Safety Act, 2021** , and projects to utilise Ravi waters flowing towards Pakistan.

What is the Punjab-Haryana and Rajasthan Water Sharing Dispute?

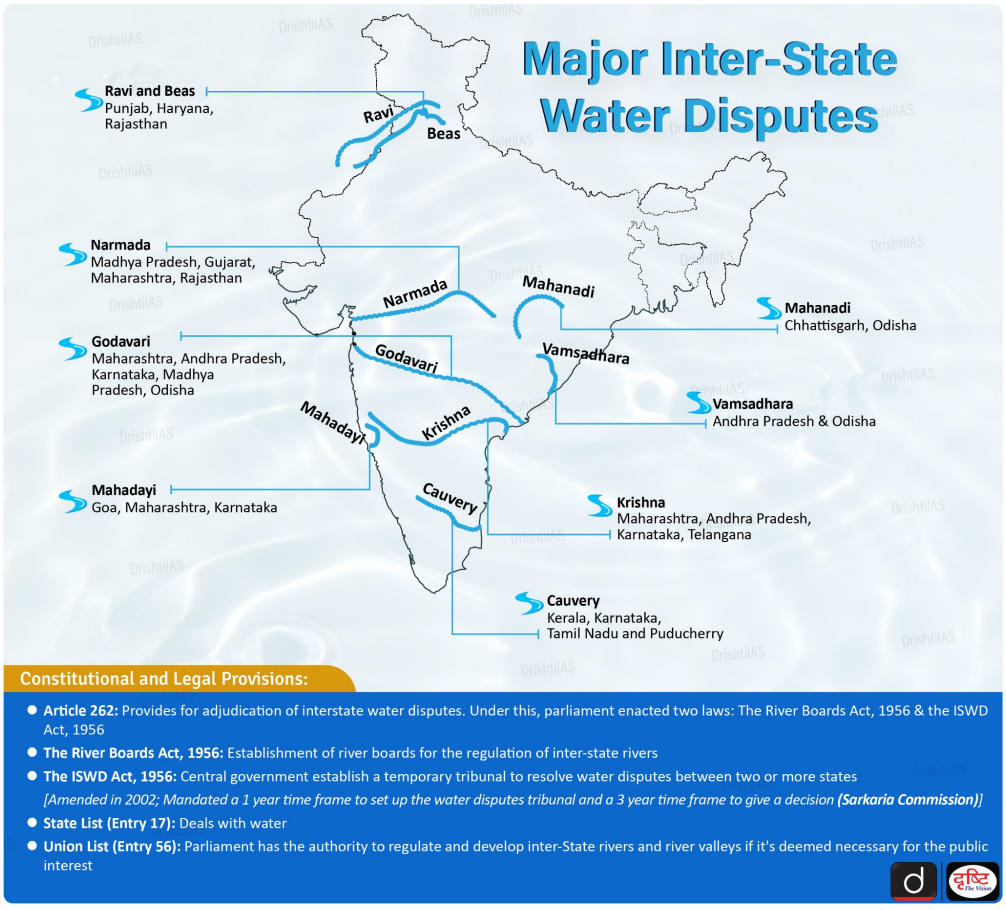
- **Initial Allocations (1955**): Under a historical arrangement dating to 1920, the princely **state of Bikaner received Sutlej waters through the Gang (Bikaner) Canal** from the Ferozepur headworks, against payments linked to water use and irrigated area.
  - In 1955, an estimated **15.85 million acre-feet (MAF) of surplus Ravi-Beas waters was allocated,** with 8 MAF to Rajasthan, 5.90 MAF to Punjab, 1.30 MAF to Patiala and East Punjab States Union (PEPSU) and 0.65 MAF to Jammu and Kashmir.
  - Following PEPSU's merger with Punjab in 1956, Punjab's combined share became 7.20 MAF.
  - The allocation was also linked to India's efforts to demonstrate planned utilisation of eastern river waters during negotiations preceding the **Indus Waters Treaty, 1960.**
- **The Indus Waters Treaty (IWT) Shift (1960):** The **IWT** granted India control over the **eastern rivers (Sutlej, Beas, Ravi) for "unrestricted use,"** enabling internal reallocation.
  - Post-1960, water sharing was treated as a **national inter-state allocation rather** than a paid commercial arrangement, **prompting Rajasthan to stop making payments to Punjab.**
- **State Reorganisation (1966):** Following Haryana's creation, the Punjab Reorganisation Act, 1966, established a framework for sharing rights and liabilities relating to the Bhakra-Nangal and Beas projects **.**
  - The Act provided for an agreement among the successor states, with the **Centre empowered to determine their respective shares if no agreement was reached** within two years.
    - Sections 79 and 80 of the act dealt with the **Bhakra Management Board and Beas Project** , respectively.
    - In 1976, the Centre allocated 3.5 MAF each to Punjab and Haryana, with 0.08 MAF recommended for Delhi.
  - The **Bhakra Beas Management Board (BBMB) ** manages key inter-state water and power infrastructure associated with the **Bhakra-Nangal and Beas projects,** supporting irrigation, hydropo **wer generation and regional water supply.**
- **The 1981 Tripartite Agreement:** A revised pact brokered by the Centre reallocated 17.17 MAF of Ravi-Beas waters.
  - Rajasthan received the largest share (8.6 MAF) (this supported the massive expansion of the **Indira Gandhi Canal**), followed by Punjab (4.22 MAF), Haryana (3.5 MAF), Jammu & Kashmir (0.65 MAF), and Delhi (0.2 MAF).
- **SYL Canal Crisis:** The pact mandated the construction of the **Sutlej-Yamuna Link (SYL) canal** to deliver Haryana’s share, sparking intense unrest and the "Nehar Roko Morcha" in Punjab in 1982.
  - The construction triggered massive political unrest and militancy in **Punjab, leaving the canal incomplete.**
- **Eradi Tribunal (1986):** Constituted following the 1985 Rajiv–Longowal Accord, the tribunal was tasked with reassessing Ravi-Beas water availability and allocations.
  - Its 1987 report recommended 5 MAF for Punjab and 3.83 MAF for Haryana, while retaining Rajasthan's 8.6 MAF allocation. However, the dispute remained unresolved.
- **Punjab Termination of Agreements Act (2004):** Punjab unilaterally enacted the Punjab Termination of Agreement Act, 2004, cancelling all its water-sharing agreements.
  - In 2016, the **Supreme Court struck down this law as unconstitutional following a Presidential Reference.** The Court stated that a state cannot unilaterally terminate binding inter-state water agreements or bypass judicial decrees regarding the SYL canal.
- **Legal Status of Assembly Resolutions:** Since the Punjab Reorganisation Act, 1966, is a Central law, Punjab cannot unilaterally repeal its provisions. It must seek amendments through Parliament or pursue appropriate judicial remedies.

Arguments of the States

- **Punjab (The Upper Riparian):** Punjab invokes the strict **Riparian Principle** , asserting that because the **Ravi, Beas, and Sutlej** do not flow through Haryana or Rajasthan, those states have no natural right to the water.
  - Punjab highlights its severe groundwater depletion crisis, claiming it has zero surplus water.
  - Furthermore, Punjab demands parity in **Yamuna waters** , arguing the Yamuna flowed through undivided Punjab pre-1966. Its recent demand for Rs 1.44 lakh crore from Rajasthan is based on a pre-1960 commercial royalty arrangement with the erstwhile Bikaner state.
  - Punjab opposes the **Dam Safety Act, 2021** because it violates the federal structure, as water is constitutionally a State subject. The state demands its repeal, fearing the Centre is using "dam safety" as a pretext to unconstitutionally seize control over Punjab's dams and irrigation headworks.
- **Haryana:** Haryana demands the immediate completion of the **SYL canal and relies strictly on the 1981 Tripartite agreement and subsequent Supreme Court decrees.**
  - The state highlights extreme arid conditions and drinking water shortages in its southern districts.
- **Rajasthan:** Rajasthan relies on the **1981 allocation of 8.6 MAF to sustain the Indira Gandhi Canal,** which is the lifeline for its desert regions.
  - It rejects Punjab's royalty demands, pointing out that post-1960 allocations are national distributions, not commercial sales.

Constitutional and Statutory Provisions for Inter-State Water Sharing

Provision	List / Act	Key Features
Entry 17	State List (List II)	Water is primarily a State subject. Covers water supplies, irrigation, canals, drainage, embankments, water storage and hydropower, subject to Entry 56 of the Union List.
Entry 56	Union List (List I)	Empowers Parliament to regulate and develop **inter-State rivers and river valleys in the public interest,** to the extent declared by law.
Entry 32	Concurrent List (List III)	Deals with shipping and navigation on inland waterways concerning mechanically propelled vessels and the rule of the road on such waterways.
Article 262	Constitution of India	Empowers Parliament to provide for the adjudication of inter-State river water disputes. Parliament may exclude the jurisdiction of the Supreme Court and other courts in such disputes.
Inter-State River Water Disputes Act, 1956	Parliamentary legislation	Enacted under **Article 262** . A State can request the Centre to refer an unresolved water dispute to a Water Disputes Tribunal. The Tribunal is to be constituted within one year of the request. Its published award has the force of an order or decree of the Supreme Court.



What Measures Can Be Taken to Address the Punjab-Haryana-Rajasthan Water Dispute?

- **Reassess Current Water Levels:** Conduct an independent, basin-wide hydrological assessment of the Ravi-Beas system, incorporating updated river flows, climate variability, environmental requirements and changing agricultural demand to support evidence-based water allocations.
- **Enforce Crop Diversification:** Urgently transition Punjab and Haryana away from water-guzzling paddy (rice) to climate-resilient crops (millets, maize, cotton) to halt severe groundwater depletion.
  - The Centre must offer guaranteed **Minimum Support Prices (MSP)** and procurement for low-water crops to financially incentivize farmers to switch.
- **Establish a Permanent Water Disputes Tribunal:** Introduce fresh legislation for a permanent inter-state river water disputes tribunal, with specialised benches, transparent hydrological assessments and time-bound adjudication to reduce prolonged litigation.
- **Promote Integrated River Basin Management:** Establish a joint River Basin Organisation (RBO) involving the concerned states to coordinate water allocation, hydrological monitoring, drought management and environmental flows.
- **Upgrade Irrigation Tech:** Massively subsidize micro-irrigation (drip and sprinkler systems) and line existing canal networks to prevent water seepage losses.
- **De-politicize the SYL Canal:** Foster cooperative federalism where the Centre acts as a neutral mediator to resolve the **Sutlej-Yamuna Link (SYL)** deadlock, moving away from vote-bank politics and endless litigation.

Conclusion

The Punjab-Haryana-Rajasthan water dispute cannot be solved by historical entitlements or rigid riparian claims alone. As India faces an impending climate crisis, the focus must shift from "who owns the water" to "how to best manage and conserve the water." A sustainable future depends on cooperative basin management and radical shifts in regional agricultural practices.

Click here to Read: **Punjab-Rajasthan Water Dispute**

Drishti Mains Question:

Water governance requires balancing riparian rights with equitable distribution and national interests. Examine in the context of India's inter-state river-water disputes.

Frequently Asked Questions (FAQs)

1. Which constitutional provision empowers Parliament to adjudicate inter-state river-water disputes?
Article 262 empowers Parliament to provide for the adjudication of inter-state river-water disputes and exclude the jurisdiction of courts in such matters.

2. What is the significance of Entry 56 of the Union List?
It empowers Parliament to regulate and develop inter-state rivers and river valleys in the public interest, to the extent declared by law.

3. What was the significance of the 1981 Ravi-Beas Water Agreement?
The agreement allocated 17.17 MAF among Punjab, Haryana, Rajasthan, Jammu and Kashmir, and Delhi, with Rajasthan receiving the largest share of 8.6 MAF.

4. What is the Sutlej-Yamuna Link (SYL) Canal dispute?
The SYL canal was proposed to transfer Punjab's allocated Ravi-Beas waters to Haryana. Its incomplete construction has remained a major source of inter-state conflict.

5. What is the role of the Inter-State River Water Disputes Act, 1956?
Enacted under Article 262, it provides for the constitution of tribunals to adjudicate inter-state river-water disputes and ensures the implementation of published tribunal awards.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/vwINxj4EQxY**]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Mains

Q. Constitutional mechanisms to resolve the inter-state water disputes have failed to address and solve the problems. Is the failure due to structural or process inadequacy or both? Discuss. **(2013)**

Tokamaks for Nuclear Fusion

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Source: TH

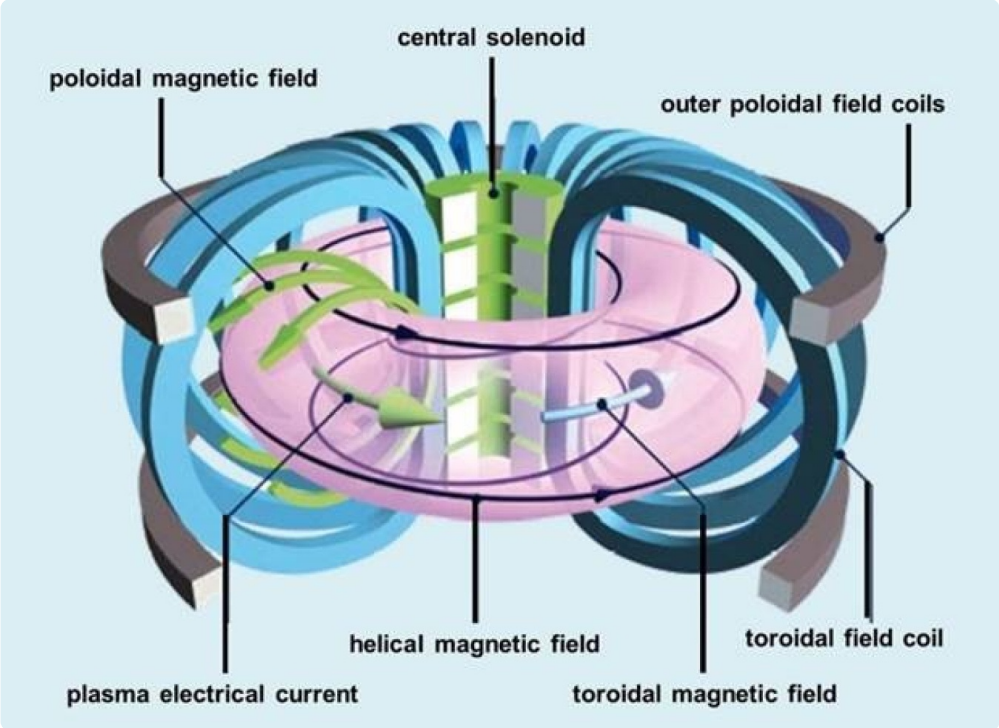
Why in News?

Bengaluru-based startup **Pranos Fusion has unveiled PRAGYA,** described as India’s first privately

developed compact **tokamak,** marking a significant step in private-sector participation in the country's **nuclear fusion research.**

What is a Tokamak?

- **Definition:** A tokamak (a Russian acronym for "toroidal chamber with magnetic coils") is an experimental machine designed to **harness fusion energy.**
- **Core Function:** Because no physical material on Earth can contain plasma heated to **150 million °C** without instantly melting, a tokamak uses a **ring-shaped (doughnut) vacuum vessel** and **powerful magnetic fields** to suspend and confine the plasma in mid-air.
- **Significance:** It is the most researched fusion reactor design globally, serving as the basis for the **International Thermonuclear Experimental Reactor (ITER)** in France, of which **India is a key member.**
- **Working of a Tokamak:**
  - **Magnetic Cage:** The tokamak creates a **highly stable "magnetic cage"** to keep the plasma away from the walls by combining two magnetic fields:
    - **Toroidal Field:** Generated by large external superconducting coils surrounding the chamber.
    - **Poloidal Field:** Generated by inducing a massive electric current directly through the plasma itself.
  - **Power Extraction:** When fusion occurs, high-energy, uncharged **neutrons** escape the magnetic cage and strike the reactor's inner wall.
  - **Tritium Breeding:** The inner wall is lined with a **Lithium blanket** . The kinetic energy of the neutrons heats the blanket (which boils water to spin electricity-generating turbines) and reacts with the lithium to breed fresh **Tritium** , ensuring the reactor sustains its own fuel supply.



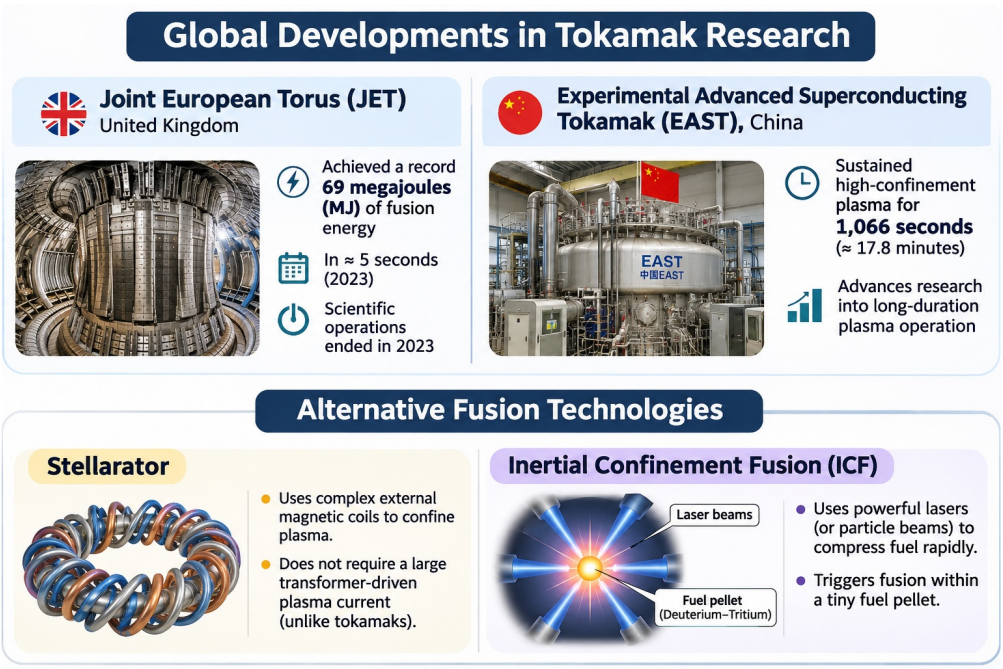
- **Advantages of Tokamak-Based Fusion:**
  - **Abundant Fuel Supply:** Deuterium can be extracted endlessly from seawater, and lithium (for Tritium breeding) is highly abundant in the Earth's crust.
  - **Zero Carbon Emissions:** The primary byproduct is harmless helium. Fusion produces absolutely **no greenhouse gases** .

- **Inherent Safety:** Unlike nuclear fission, a tokamak cannot undergo a runaway chain reaction or meltdown. Any disruption to the magnetic cage immediately cools the plasma, halting the reaction in seconds.
- **No Long-Lived Waste:** It does not produce the highly radioactive, long-lasting nuclear waste associated with traditional uranium-based fission reactors.

- **Limitations:**
  - **Net Energy Deficit (Q-Value):** Maintaining the plasma requires massive amounts of power. **No tokamak in the world has yet generated a net energy gain for commercial electricity (a Q-value > 1).**
  - **Material Degradation:** The reactor walls face extreme, continuous bombardment by high-energy neutrons, requiring the development of advanced radiation-resistant materials.
  - **Tritium Scarcity:** Tritium is highly radioactive and naturally rare; commercial viability depends entirely on the unproven efficiency of the lithium blanket breeding technology.

India's Status on Tokamaks

- **Public Sector Leadership** : The **Institute for Plasma Research (IPR)** in Gandhinagar, Gujarat, an autonomous institution under the Department of Atomic Energy (DAE), leads India's public-sector plasma and fusion research.
  - It operates indigenous experimental tokamaks, **including ADITYA-U (upgraded from ADITYA, India's first tokamak) and the Steady State Superconducting Tokamak (SST-1).**
- **Global Collaboration:** The ITER project in France aims to generate 500 MW of fusion power from **50 MW of external plasma-heating power (Q ≥ 10). However, ITER will not generate electricity, as it is an experimental facility.**
  - **India is one of the seven ITER members** , alongside China, the European Union, Japan, South Korea, Russia and the United States, contributing approximately 9% of the project's construction value and supplying critical components such as the cryostat.
- **Private Sector Emergence:** The recent unveiling of **PRAGYA** by Bengaluru-based **Pranos Fusion** marks a critical shift. PRAGYA is a compact, low-aspect-ratio tokamak designed to test advanced plasma control and High-Temperature Superconducting (HTS) magnets, positioning India's private sector in the global race for commercial fusion.



Nuclear Fusion

- **Definition:** Nuclear fusion is the fundamental **physical process that powers the Sun and stars** . It occurs when two light atomic nuclei- (most commonly the hydrogen isotopes **Deuterium and Tritium - collide** at extreme velocities and fuse together to create a heavier nucleus (Helium).
- **Energy Generation:** The resulting Helium atom has slightly less mass than the original hydrogen atoms combined.
  - This tiny difference in mass, known scientifically as the **mass defect** , is converted into a massive amount of **clean energy** , governed by **Albert Einstein's mass-energy equivalence equation, $E = mc^2$.**
- **The Plasma State Condition:** To overcome the natural electrostatic repulsion between positively charged nuclei, the fusion fuel must be heated to extreme temperatures exceeding **150 million °C** .
  - Under this intense heat, the gas transforms into **plasma** (a superheated **fourth state of matter)** consisting of a highly energetic soup of free-roaming electrons and charged ions.

Frequently Asked Questions (FAQs)

1. What is nuclear fusion and how does it generate energy?
Nuclear fusion occurs when light atomic nuclei, such as deuterium and tritium, combine to form a heavier nucleus. The resulting mass defect is converted into energy according to $E = mc^2$.

2. What is a tokamak and how does it confine plasma?
A tokamak is a doughnut-shaped fusion device that uses toroidal and poloidal magnetic fields to confine extremely hot plasma without allowing it to touch the reactor walls.

3. What is the significance of the ITER project?
ITER, located in France, is an international nuclear fusion research project involving India and six other members. It aims to demonstrate the scientific and technological feasibility of fusion energy.

4. What are the major challenges in achieving commercial nuclear fusion?
Key challenges include achieving sustained net energy gain, maintaining plasma stability, developing radiation-resistant materials and ensuring efficient tritium breeding.

5. What are India's major initiatives in tokamak research?
The Institute for Plasma Research (IPR) operates ADITYA-U and SST-1. India's participation in ITER and the emergence of private initiatives such as PRAGYA are expanding the domestic fusion research ecosystem.

[Watch Video on YouTube: ▶ **https://www.youtube.com/embed/0iFFaDDSeBk**]

UPSC Civil Services Examination, Previous Year Questions (PYQs)

Prelims

Q. The function of heavy water in a nuclear reactor is to (2011)

(a) Slow down the speed of neutrons

(b) Increase the speed of neutrons

(c) Cool down the reactor

(d) Stop the nuclear reaction

Ans: (a)

Mains

Q. With growing energy needs should India keep on expanding its nuclear energy programme? Discuss the facts and fears associated with nuclear energy. **(2018)**

India Targets 1st Human Deep-Sea Mission to 6,000 Metres by 2028

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Source: DD

India aims to undertake its **first human deep-sea mission to a depth of up to 6,000 metres** by 2028 under the **Samudrayaan Mission** , as highlighted at the **Indo-Pacific Regional Dialogue (IPRD) 2026** , emphasising indigenous technological capabilities and international scientific cooperation.

- **Samudrayaan Mission** : India's human deep-sea exploration initiative under the **Deep Ocean Mission ,** aimed at developing indigenous technologies for deep-sea exploration and scientific research.
  - The Deep Ocean Mission, approved by the **Union Cabinet in 2021** and implemented by the **Ministry of Earth Sciences** , focuses on deep-sea mining technologies, ocean climate services, marine biodiversity conservation, seabed exploration, and ocean-based energy and freshwater resources.
- **MATSYA-6000 Submersible:** An **indigenously developed human-occupied,** battery-powered submersible, designed by the **National Institute of Ocean Technology (NIOT), Chennai,** to carry three aquanauts to depths of up to 6,000 metres.
  - It is designed for **12 hours of normal operation, with 96 hours of emergency life-support capability,** and incorporates a titanium-alloy personnel sphere to withstand extreme underwater pressure.
- **Ocean Mineral Explorer:** Supports **high-resolution exploration of hydrothermal sulphide** deposits on the seabed, contributing to the identification and assessment of strategic deep-sea mineral resources.
- **Space–Ocean Technology Convergence:** Satellite observations of ocean colour, sea-surface temperature and wind vectors strengthen ocean monitoring, climate research and assessment of marine **renewable energy** potential.
- **Applications:** Facilitates direct observation and scientific study of deep-sea biodiversity, seabed geology and extreme marine environments, improving understanding of largely unexplored ocean ecosystems.
- **International Cooperation:** India has proposed joint scientific missions, reciprocal access to research facilities and data sharing under the principle of **"One Ocean, Shared Knowledge and Equitable Prosperity."**
  - Key regional priorities include improving **ocean observation networks, extreme-event forecasting, coastal climate services and sustainable management of marine resources.**
- **Significance:** Strengthens India's indigenous deep-sea technological capabilities, strategic resource security and interdisciplinary scientific research, while supporting a sustainable **Blue Economy** and the vision of **Viksit Bharat 2047.**

Read more: **Samudrayaan Project**

RBI Constitutes Technical Consultative Committee for Financial Markets

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Source: TH

Recently, the **Reserve Bank of India (RBI)** constituted the **Technical Consultative Committee for Financial Markets (TCCFM)** to strengthen consultation with market participants on regulatory, operational and policy issues affecting India's financial markets.

- **Chairperson:** The committee is chaired by **RBI Deputy Governor Rohit Jain** , who is in charge of the **Financial Markets Regulation Department (FMRD).**
- **Composition** : The **eight-member committee** comprises the RBI Deputy Governor and seven external members representing banks, market infrastructure institutions, financial research organisations and securities markets **.**
- **Areas of Focus:** Money markets, government securities markets, foreign exchange markets and their associated derivatives markets.
- **Market Infrastructure:** Examines **trading, clearing, settlement and reporting mechanisms** to improve market efficiency and resilience.
- **Market Deepening:** Identifies measures to deepen and broaden financial markets, improving their functioning, liquidity, efficiency and resilience.
- **Emerging Risks:** Reviews **domestic and international developments, financial innovations** and emerging risks affecting financial markets.
- **Consultative Mechanism:** Provides a **platform for market participants to highlight operational difficulties, regulatory challenges** and unintended market frictions arising from regulatory measures.
- **Significance:** Strengthens **regulatory consultation, improves financial market liquidity and efficiency, promotes robust market infrastructure** and enhances India's capacity to respond to emerging financial risks **.**

Read more: **Reserve Bank of India**

National Tabletop Exercise ‘VAJRA-SAT’

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Source: PIB

Recently, the **Department of Animal Husbandry and Dairying (DAHD), in collaboration with ICAR-National Institute of Veterinary Epidemiology and Disease Informatics(ICAR-NIVEDI),** conducted a **two-day National Tabletop Exercise ‘VAJRA-SAT’** on preparedness against **Foot-and-Mouth Disease (FMD) serotype SAT-1** in Bengaluru, Karnataka.

- **Objective:** Strengthen preparedness against the **potential entry of FMD SAT-1** , which has not been reported in India, following its geographical spread beyond Africa to parts of West Asia.
- **Theme:** "Detect Early, Confirm Fast, Contain Together".
- **Simulated Scenarios:** Assessed **disease detection, laboratory diagnosis, epidemiological investigations, animal movement restrictions, emergency vaccination and risk communication.**
- **Surveillance:** Emphasised **risk-based surveillance, epidemiological intelligence and early-warning systems.**
- **Institutional Preparedness:** DAHD proposed strengthening regional FMD laboratories and Regional Disease Diagnostic Laboratories (RDDLs) under the **Pandemic Fund,** with technical support from ICAR-National Institute of Foot-and-Mouth Disease (ICAR-NIFMD).
- **Vaccine Preparedness:** Highlighted the need for **indigenous SAT-1 vaccine development and advanced vaccine readiness,** as existing FMD vaccines used in India do not protect against SAT-1.

Foot-and-Mouth Disease (FMD)

- **About:** A highly contagious viral disease of **cloven-hoofed animals (cattle, buffaloes, sheep, goats and pigs),** caused by **FMDV (genus Aphthovirus, family Picornaviridae).**
- **Serotypes:** Seven serotypes— **O, A, C, SAT-1, SAT-2, SAT-3 and Asia-1. Of these, O, A and Asia-1 occur in India** ; SAT serotypes have not been reported, while serotype C has not been detected globally since 2004. Serotype-specific vaccination is essential.
- **Transmission and Symptoms:** Spreads through **direct contact, contaminated materials and airborne transmission** , causing fever, blisters in the mouth, feet and teats, and reduced milk production.

Read more: **Schemes to Control FMD and Brucellosis - Drishti IAS**

Indian Navy to Get 8 Next-Generation Corvettes

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Source: IE

The **Ministry of Defence (MoD)** signed contracts worth **₹53,866.50 crore** with **Garden Reach Shipbuilders & Engineers (GRSE)** and **Goa Shipyard Limited (GSL)** for the construction of **eight Next-Generation Corvettes (NGCs)** for the **Indian Navy** , following approval by the **Cabinet Committee on Security (CCS)** .

- **Procurement and Shipbuilders:** The **Defence Acquisition Council (DAC)** granted **Acceptance of Necessity (AoN)** in 2022. Under the contracts signed in 2026, **GRSE, Kolkata, will construct five corvettes** , while **GSL, Goa, will build three** .
- **Indigenous Design:** The corvettes will be **designed by the Indian Navy** and constructed by **Defence Public Sector Undertakings (DPSUs) selected through competitive bidding** , marking a significant milestone in indigenous warship construction.
  - A **corvette** is a relatively small surface combatant, generally smaller than a frigate or destroyer.
- **Operational Capabilities:** Equipped with **advanced missiles, weapons and sensor suites** , the corvettes will undertake **anti-air, anti-surface and anti-submarine warfare (ASW)** , maritime surveillance, coastal defence, littoral combat and protection of maritime trade routes.
- **Stealth and Specifications:** The warships will feature **stealth capabilities and advanced propulsion systems** . GRSE's reported specifications indicate a length of **127.4 metres** , a displacement figure of approximately **4,950 tonnes** and a maximum speed of **26 knots** .
- **Existing Fleet:** The new corvettes will complement the Navy's **four Kamorta-class corvettes** , which are primarily designed for **anti-submarine warfare** .
- **Significance:** The project will strengthen India's **maritime security and surveillance capabilities in the Indian Ocean Region** , reduce operational pressure on larger warships and promote **Aatmanirbhar Bharat** through indigenous defence manufacturing, MSME participation and employment generation.

Read more: Mahe: Indigenous Anti-submarine Warfare Vessel

Amazon Rainforest No Longer a Carbon Sink

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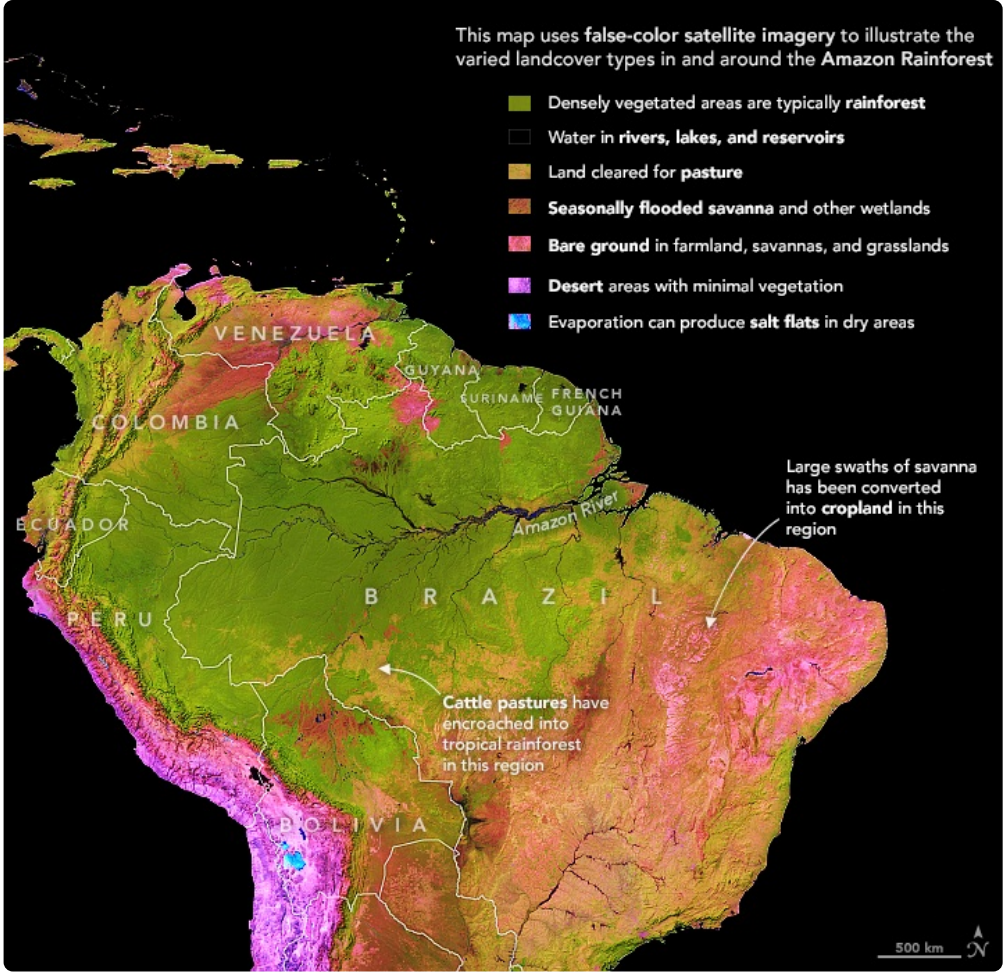
Source: TH

Recently, a study published in the journal Nature found that the **Amazon rainforest** has become **a net source of carbon emissions rather than a carbon sink** , **as deforestation, forest fires and rising temperatures** have weakened its capacity to absorb carbon dioxide.

- **Carbon Emissions:** The Amazon emitted an annual average of 1.17 billion tonnes of CO_2 more than it absorbed, with **forest fires** alone releasing approximately 1.47 billion tonnes of CO_2 annually.
  - 2024 recorded the highest annual net emissions, approximately 2.97 billion tonnes of CO_2, during the study period.
- **Declining Rainfall:** Precipitation declined by **17% during the dry-season peak (August–October) between 1979 and 2018, alongside a 0.9°C rise** in annual temperature.
- **Forest Fires and El Niño: El Niño-induced droughts** and rising temperatures intensified forest fires. Only 1.6% and 1.0% of fire events were associated with lightning in 2023 and 2024, respectively, indicating that **most fires were human-induced and aggravated by drought conditions.**
- **Human Activities:** Expansion of **mining, pastureland and soybean plantation** s, along with deforestation and **biomass burning** , are major drivers of forest degradation.
- **Implication:** The weakening of the Amazon's carbon sink **threatens global climate regulation, accelerates atmospheric CO_2 accumulation** and underscores the need for zero-deforestation policies, fire prevention and forest restoration.

Amazon Rainforest

- **Location and Area:** Spans **eight South American countries and French Guiana** , with the **largest portion in Brazil** . Covering approximately 5.5 million sq. km, it is the world's largest tropical rainforest.
- **Biodiversity and Climate** : Supports approximately **10% of the world's known species** and regulates carbon storage, moisture recycling and regional rainfall.
- **River and Livelihoods:** The **Amazon River** has the world's highest water discharge, while the wider Amazon region supports over 40 million people, including Indigenous communities.



Read more: Amazon Rainforest